



REAL-TIME STRESS DETECTION SYSTEM

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CERTIFICATE

This is to certify that the project work titled “Real-time Stress Detection System” is carried out by Haritha A S (1MS23EC042), Syed Maaz (1MS23EC139), Vidit Agrawal (1MS23EC144), Yash Kumar (1MS23EC150), bonafide students of Ramaiah Institute of Technology, Bangalore, as part of Mini-project work carried out in sixth semester of Bachelor of Engineering in Electronics and Communication during the year 2025-2026. It is certified that all corrections / suggestions indicated for Internal Assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements prescribed.

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DECLARATION

We hereby declare that the mini-project entitled “Real-time Stress Detection System” has been carried out independently at Ramaiah Institute of Technology under the guidance of Dr. Lakshmi S, Professor, Department of Electronics and Communication Engineering, RIT, Bangalore.

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ABSTRACT

Stress is one of the most widespread health concerns of modern life, yet it is also one of the least visually apparent. Despite rapid advancements in wearable health technology, important physiological indicators of human stress — such as Heart Rate Variability (HRV) and Galvanic Skin Response (GSR) — remain mostly invisible and difficult for users to interpret in daily life. Current biometric systems present this data as complex numerical readings and clinical dashboards, which do not help users build an immediate emotional connection or intuitive understanding of their internal physiological state. This project presents a Real-time Stress Detection System that bridges this gap using non-invasive physiological sensors, an embedded microcontroller, and generative visual media tools. The system captures HRV and GSR signals from the user's body using a MAX30102 pulse sensor and a GSR sensor interfaced with an ESP32 microcontroller. Processed sensor data is transmitted via a Python serial-to-OSC bridge to TouchDesigner, where it is mapped to color-coded visual representations overlaid on the user's live silhouette captured via webcam. The system was successfully implemented and tested, demonstrating end-to-end integration from biosensor data acquisition to real-time visual output at 60 FPS. The colored silhouette updates dynamically — rendering green for calm/normal states and red for stressed states — creating an immersive and intuitive biofeedback experience. The system demonstrates the feasibility of integrating embedded biosensors with generative media software to produce ambient, real-time biofeedback that is both meaningful and accessible to everyday users.

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List of Acronyms

Acronym	Full Form
HRV	Heart Rate Variability
GSR	Galvanic Skin Response
ESP32	Espressif Systems 32-bit Microcontroller
OSC	Open Sound Control
BPM	Beats Per Minute
RMSSD	Root Mean Square of Successive Differences
PPG	Photoplethysmography
IoT	Internet of Things
UDP	User Datagram Protocol
OLED	Organic Light Emitting Diode
I2C	Inter-Integrated Circuit
ADC	Analog-to-Digital Converter
CHOP	Channel Operator (TouchDesigner)
TOP	Texture Operator (TouchDesigner)
FPS	Frames Per Second
IR	Infrared
RGB	Red Green Blue

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CHAPTER 1

INTRODUCTION AND PROBLEM STATEMENT

1.1 Introduction

In everyday life, stress often goes unnoticed and unaddressed. Mothers, teachers, employees, doctors, nurses, and other working individuals may experience constant pressure and assume it is simply part of their routine, ignoring prolonged fatigue and tension as normal. Even when the body shows signs such as increased heart rate, headaches, or restlessness, these signals are often overlooked until they develop into burnout or more serious health issues.

Stress is a fundamental physiological and psychological response to demanding or challenging environmental situations. While temporary stress can be a motivator, chronic exposure to high stress levels is a severe health risk, often leading to cardiovascular disorders, chronic anxiety, metabolic diseases, and a significant reduction in cognitive and motor performance. Traditionally, stress assessment has relied on clinical evaluations and subjective self-reported questionnaires. These methods are inherently flawed because they are subjective, dependent on infrequent hospital visits, and unsuitable for continuous, real-time monitoring required to capture stress as it occurs in daily life.

Recent advances in Internet of Things (IoT) and wearable biosensor technology have enabled continuous and objective stress monitoring through physiological markers such as Heart Rate Variability (HRV) and Galvanic Skin Response (GSR). While modern smartwatches and fitness bands provide health metrics like heart rate and sleep data, they often present information primarily for tracking rather than real-time awareness. To address this gap, the proposed system transforms physiological signals into intuitive visual feedback, creating a real-time generative artwork that reflects the user's stress level and promotes greater awareness of their physiological state.

The Real-Time Stress Detection System proposed in this project is designed to bridge this gap using non-invasive physiological sensors, embedded microcontrollers, and generative visual media tools. By capturing Heart Rate Variability (HRV) and Galvanic Skin Response (GSR) signals from the user's body and mapping them to color-coded visual representations in real time, the system translates invisible biological data into intuitive visual feedback.

The system utilizes an ESP32 microcontroller for data acquisition and processing, Python for serial-to-OSC communication, and TouchDesigner for generating the visual output. A webcam feed is used to extract the user's silhouette, which is dynamically colored according to the detected stress level. This creates an immersive biofeedback experience that allows users to become more aware of their physiological state and encourages proactive stress management through intuitive visual interaction.

1.2 Problem Statement

Despite significant advancements in wearable health technology, physiological indicators of stress such as Heart Rate Variability (HRV) and Galvanic Skin Response (GSR) remain difficult for users to interpret in everyday life. Most existing systems present this information through numerical readings, dashboards, and graphs, making it challenging for users to intuitively understand their stress levels, therefore, there is a need for a non-invasive and ambient biofeedback system that transforms complex physiological signals into intuitive visual feedback. Such a system can help users recognize, understand, and potentially regulate their stress levels in real time.

The key gaps in current technology are:

- **Lack of Real-Time Intervention:** Most wearable devices act as passive monitoring tools, collecting physiological data without providing immediate feedback to help users recognize and manage stress as it occurs.
- **Latency and Privacy Issues:** Many existing systems rely on cloud-based processing, which can introduce delays, increase power consumption, and raise concerns regarding the privacy of sensitive physiological data.
- **Classification Challenges:** Distinguishing stress-related physiological changes from those caused by physical activity remains a major challenge, often reducing the accuracy of stress detection systems.

1.3 Objectives and Scope

1.3.1 Objective

The primary objectives of this work are:

1. To design and implement a non-invasive, real-time stress detection system using biosensors.
2. To acquire and process HRV (Heart Rate Variability) and GSR (Galvanic Skin Response) signals using an ESP32 microcontroller.
3. To transmit processed sensor data from Python to TouchDesigner using OSC (Open Sound Control) protocol.
4. To develop a real-time generative visual system that maps physiological stress levels to intuitive color states overlaid on a user's live silhouette.
5. To demonstrate the feasibility of integrating embedded biosensors with generative media systems for interactive biofeedback applications.

1.3.2 Scope

The scope of this project is defined in direct correspondence with the objectives stated above:

- **Sensor coverage:** The system acquires two non-invasive physiological signals — heart rate variability via the MAX30102 optical pulse sensor and skin conductance via a GSR sensor. Clinical-grade accuracy and medical diagnosis are outside the scope of this work.
- **On-device processing:** All signal filtering, beat detection, RMSSD-based HRV computation, and GSR oversampling are performed on the ESP32 microcontroller. The processed values are transmitted to the host machine over USB serial at a 500 ms update rate.
- **Data transmission:** A Python middleware layer receives the serial data, performs baseline calibration and state classification, and forwards the result to TouchDesigner using the OSC protocol over UDP. The system operates on a local network and does not involve cloud connectivity or remote data storage.
- **Visual output:** The classified emotional state is rendered as a coloured body silhouette on a fullscreen projection surface in TouchDesigner. Four states are supported — *Zen*, *Stress*, *Flow*, and *Burnout* — for a single user at a time.
- **Integration feasibility:** The complete pipeline from sensor acquisition on embedded hardware to generative visual output is implemented and validated as a proof-of-concept installation. Multi-user operation, wireless sensing, and machine learning classification are identified as directions for future work.

CHAPTER 2

LITERATURE SURVEY

2.1 “Design of an IoT-enabled Wearable Device for Stress Level Monitoring,” International Journal of Innovative Research and Scientific Studies, vol. 8, no. 1, pp. 599–612, Jan. 2025.

This paper presents an IoT-enabled wearable platform that integrates MAX30102 (PPG) and GSR sensors for real-time stress monitoring. Physiological data from multiple sensors are collected and transmitted through Wi-Fi to a Firebase cloud platform for storage and visualization. The study emphasizes the importance of low-cost embedded hardware for portable healthcare applications and investigates the impact of continuous cloud communication on system performance and battery life. The results demonstrate that wearable devices can effectively monitor stress levels while highlighting challenges related to power efficiency and long-term operation.

Takeaway: The study validated the use of MAX30102 and GSR sensors for real-time stress detection. It also showed that continuous cloud transmission increases power consumption, motivating the use of edge processing in the proposed system to improve efficiency and reduce latency.

2.2 "IoT-Based Stress Monitoring Using CNN for HRV-GSR Analysis,” Sinkron: Jurnal dan Penelitian Teknik Informatika, vol. 10, no. 1, pp. 621–637, 2026.

This paper presents an advanced framework for stress monitoring that leverages the fusion of Heart Rate Variability (HRV) and Galvanic Skin Response (GSR) signals processed through Convolutional Neural Networks (CNN). By integrating multi-modal biosignals, the researchers demonstrate a significant improvement in classification accuracy, effectively overcoming the limitations of single-sensor systems. The study emphasizes the computational potential of deep learning architectures in embedded IoT environments to interpret complex physiological patterns. Furthermore, the findings indicate that extracting spatial features from time-series biosignal data via CNNs provides a more robust indicator of psychological stress than traditional threshold-based detection methods, thereby enhancing the reliability of the diagnostic output.

Takeaway: The research confirms that multi-modal sensor fusion combined with deep learning significantly boosts classification precision in stress detection. It demonstrates that CNNs are highly effective for feature extraction from raw physiological data, which directly supports the potential for implementing lightweight machine learning models on local hardware to enhance overall system intelligence.

2.3 “Stress Detection Using Bio-Signal Processing: An Application of IoT and Machine Learning,” Engineering Proceedings, vol. 81, no. 1, 2025.

This paper utilizes an integrated framework of IoT infrastructure and machine learning algorithms to facilitate stress detection specifically for residents in old-age homes. The study addresses the challenges of deploying sophisticated bio-signal processing systems within actual care environments, ensuring that the technology is both practical and effective in real-world settings. By synthesizing physiological sensor data with robust classification models, the authors demonstrate a scalable method for continuous health monitoring. The research underscores the necessity of adapting diagnostic tools to meet the specific requirements of elderly users, thereby enhancing the quality of care through proactive stress management and non-invasive monitoring techniques.

Takeaway: The deployment proves that combining physiological sensor data with machine learning is highly effective for stress monitoring in sensitive, real-world environments. It establishes a strong precedent for using automated classification to provide consistent monitoring, confirming that such systems can be reliably applied to improve health outcomes in assisted living contexts.

2.4 “IoT-Based Stress and Non-Invasive Glucose Monitoring System,” H. Khadar et al., RV College of Engineering, May 2025.

This paper introduces an integrated IoT-based framework designed for the simultaneous, non-invasive monitoring of stress levels and glucose concentrations. By consolidating these distinct health parameters into a single device, the research demonstrates the technical feasibility of utilizing compact embedded systems to facilitate continuous, comprehensive physiological tracking. The study highlights significant advancements in sensor miniaturization and data aggregation, providing a viable pathway for more convenient, long-term health management solutions. Furthermore, the architecture emphasizes the utility of multi-parameter monitoring for early detection of health trends, contributing valuable insights into the development of holistic wearable technology for diverse user needs.

Takeaway: The research successfully demonstrates that multiple health metrics can be reliably tracked simultaneously using compact embedded hardware. It validates the potential for high-efficiency, multi-modal wearable systems to replace single-purpose devices, offering a scalable approach for integrated, continuous patient monitoring in real-world applications.

2.5 “Non-Invasive Monitoring of Blood Glucose Using Sensors and AI,” Sensors and Materials, vol. 32, no. 7, pp. 2545–2556, 2020.

This paper provides a foundational investigation into the methodologies for non-invasive physiological parameter monitoring, specifically focusing on blood glucose estimation through a combination of optical sensing and artificial intelligence. By exploring the complex relationship between non-contact biosensing signals and blood chemistry, the research establishes essential protocols for mitigating environmental noise and signal interference. The authors demonstrate that AI-driven signal processing techniques can effectively correlate sensor-derived data with blood glucose levels, thereby laying the

groundwork for more sophisticated, wearable diagnostic platforms. This study serves as a critical reference point for understanding the current limitations and future potential of non-invasive sensing in clinical and personal health applications.

Takeaway: The study confirms that AI-enhanced signal processing is essential for achieving accurate results in non-invasive glucose monitoring. It underscores that foundational advancements in sensor design and data interpretation are key to transitioning from invasive sampling to reliable, contact-less monitoring systems.

2.6 “Design of Stress Detector with Fuzzy Logic Method — GSR and Heart Rate Parameters,” Indonesian Journal of Electrical Engineering and Computer Science, vol. 37, no. 1, pp. 56–68, 2025.

This paper presents a specialized stress detection architecture that utilizes fuzzy logic to integrate disparate Galvanic Skin Response (GSR) and heart rate parameters. By employing a multi-level fuzzy inference system, the authors effectively classify physiological data into granular stress levels, providing a more nuanced analysis than traditional binary detection systems. The study explores the mathematical modeling of human physiological responses to emotional stimuli, demonstrating how fuzzy membership functions can manage the inherent uncertainty and non-linearity of sensor data. This methodology offers a robust alternative to conventional threshold-based systems by allowing for dynamic decision-making processes that better mirror biological variability in stress manifestation.

Takeaway: The research confirms that fuzzy logic is highly effective for interpreting complex multi-modal physiological data and provides a superior method for multi-level stress classification. This approach directly informs the refinement of the threshold-based classification logic implemented in this project, enabling a more adaptive and accurate response to varying physiological inputs.

2.7 “Stress Detection Based on ECG, GSR Signals, HR, and Behavioral Context,” Power System Technology, vol. 49, no. 2, 2025.

This paper introduces a comprehensive stress detection framework that synthesizes electrocardiogram (ECG) data, Galvanic Skin Response (GSR), heart rate monitoring, and behavioral contextual analysis. By incorporating environmental and activity-based data alongside physiological signals, the research successfully mitigates false positives that often arise from physical exertion rather than psychological stress. The authors establish that contextual awareness is critical for differentiating between genuine emotional distress and natural variations in biometrics caused by daily movement. This multi-layered approach provides a roadmap for increasing the reliability of wearable health devices by ensuring that systemic decisions are supported by both internal biological state and external situational evidence.

Takeaway: The study demonstrates that integrating behavioral context with physiological data is essential for achieving high-accuracy stress classification. It reinforces the validity of a multi-signal approach and emphasizes that incorporating situational awareness significantly improves the system’s ability to discern genuine stress from environmental noise.

2.8 “Stress Detection Using Low Cost Heart Rate Sensors,” *Sensors*, vol. 16, no. 10, 2016.

This foundational study demonstrates the feasibility of achieving accurate stress detection utilizing low-cost heart rate sensors as the primary input modality. By systematically evaluating the correlation between heart rate variability and psychological stress states, the authors establish that high-fidelity monitoring can be accomplished without the necessity for expensive or intrusive clinical-grade equipment. The research provides a rigorous validation of affordable biosensor architectures, highlighting their potential for widespread deployment in personal health monitoring applications. Furthermore, this investigation underscores that when signal processing algorithms are appropriately optimized, even entry-level hardware can deliver reliable diagnostic insights, thereby effectively lowering the barrier to entry for developing community-accessible stress detection technologies.

Takeaway: The research validates the viability of utilizing affordable, low-cost biosensor systems for effective stress monitoring. This finding directly informed the hardware selection strategy for this project, confirming that cost-effective sensor components can meet the necessary performance benchmarks for real-world stress classification.

2.9 “Internet of Tangible Things (IoTT): Challenges and Opportunities for Tangible Interaction with IoT,” *Informatics*, vol. 5, no. 1, p. 7, 2018.

This paper introduces the paradigm of the Internet of Tangible Things (IoTT), which pivots from traditional device-centric connectivity toward human-centered interaction facilitated by physical objects. By prioritizing experience-driven design, the study explores how real-time feedback loops can bridge the gap between human intent and machine response. The authors emphasize the importance of continuous sensor data acquisition and the development of intuitive mappings between physical interactions and digital system outputs. This conceptual framework provides a robust foundation for implementing biofeedback visualization, as it validates the necessity for high-frequency data integration to ensure that the interface feels responsive and meaningful to the user.

Takeaway: The research highlights that successful IoT integration requires designing for human-centered interaction rather than mere connectivity. This approach directly supports the TouchDesigner-based biofeedback visualization strategy, reinforcing that real-time mapping of physiological data to visual outputs is critical for creating an immersive and functional tangible interface.

2.10 “Whispers of the Fan: An Interactive Installation Artwork Based on Arduino Sensors Integrated with TouchDesigner,” *Smart Innovation, Systems and Technologies*, vol. 429, pp. 601–618, 2025.

This paper investigates the seamless integration of Arduino-based physical sensors with TouchDesigner to facilitate the creation of real-time interactive visual art installations. The authors analyze the technical requirements for bridging hardware sensor inputs with creative software environments, ensuring low-latency data transmission for dynamic artistic output. By exploring the aesthetic and functional

possibilities of interactive installations, the study highlights how physical actions can be translated into immersive visual experiences. This research provides a technical blueprint for configuring communicative interfaces that respond instantaneously to external stimuli, thereby expanding the potential for interactive design in both public and private spaces.

Takeaway: The research confirms that real-time sensor data must be meaningfully mapped into immediate visual feedback to create effective interactive experiences. This principle directly inspired the generative visual biofeedback design utilized in this project, ensuring that physiological data is represented through responsive and engaging digital visual elements.

CHAPTER 3

DESIGN AND METHODOLOGY

3.1 Design

The proposed Real-Time Stress Detection System is designed as a multi-stage pipeline consisting of physiological sensing, signal processing, emotion classification, and visual biofeedback, as shown in Figure 3.1. The system combines embedded hardware, Python-based middleware, and TouchDesigner to convert physiological signals into an intuitive visual representation of the user's emotional state.

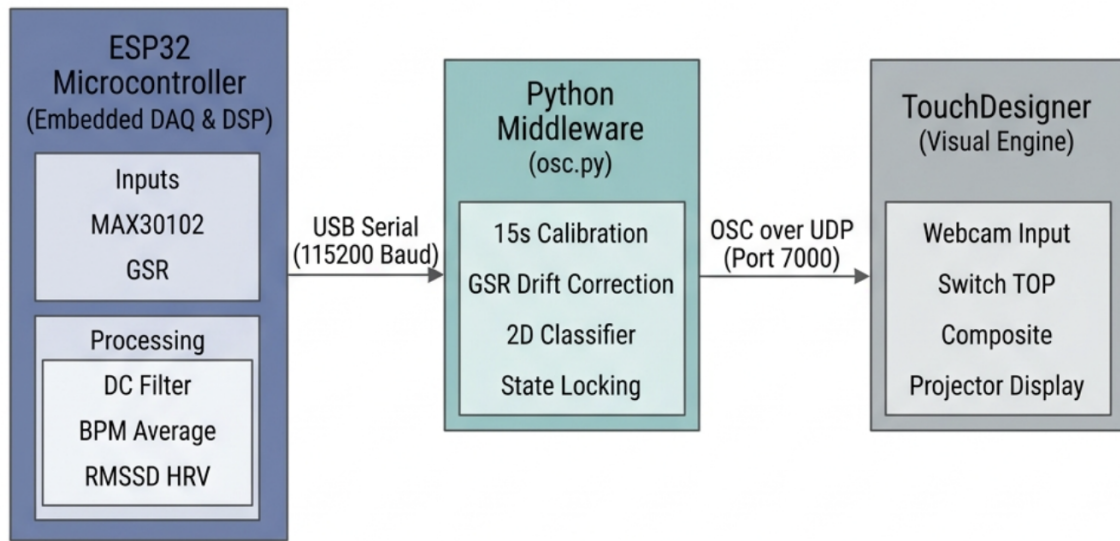


Figure 3.1: System Design Overview

The first stage of the system is the **sensor layer**, where physiological data is collected from the user. A finger is placed on a sensing module containing a MAX30102 pulse sensor and a Galvanic Skin Response (GSR) sensor. The MAX30102 measures pulse-related blood volume changes using optical sensing, while the GSR sensor measures changes in skin conductance caused by variations in sweat gland activity. These two signals provide information about the user's cardiovascular activity and emotional arousal.

The acquired sensor data is processed by the **ESP32 microcontroller**, which acts as the primary data acquisition and signal processing unit. Since the raw sensor signals contain noise and unwanted variations, the ESP32 applies a DC Filter to stabilize the pulse waveform and remove slow signal drift. The controller then calculates the user's Heart Rate (BPM) by averaging the time between consecutive heartbeats. In addition, Heart Rate Variability (HRV) is computed using the RMSSD (Root Mean Square of Successive Differences) method, which measures variations between heartbeat intervals. The GSR readings are also averaged to reduce electrical noise and improve signal stability.

After processing, the ESP32 packages the physiological parameters into a text-based data packet and

transmits them to the host computer through a USB connection using the UART protocol at 115200 bps. The data is updated every 500 ms, enabling continuous monitoring of the user's physiological state.

The **software processing stage** is handled by a Python middleware application (`osc.py`), which calibrates a personal HRV and GSR baseline in the first 15 seconds, applies GSR drift correction, and maps readings onto the 2D Circumplex Model of Affect to classify the user's state as *Zen*, *Stress*, *Flow*, or *Burnout*. A three-reading confirmation gate prevents false state transitions.

The final stage is the **visual biofeedback module** in TouchDesigner, which receives the classified state via OSC over UDP 7000, extracts a live silhouette from the webcam feed, and fills it with the corresponding colour — Violet, Hot Pink, Cyan, or Teal. Removing the finger clears the display, completing the closed-loop biofeedback system.

3.2 Methodology

The flowchart gives an overview of the entire project and is explained further in this chapter as three different parts:

1. Hardware Sensing and DSP (ESP32)
2. Middleware & Emotion Classification (Python: `osc.py`)
3. TouchDesigner Setup 2.0 Environment Architecture

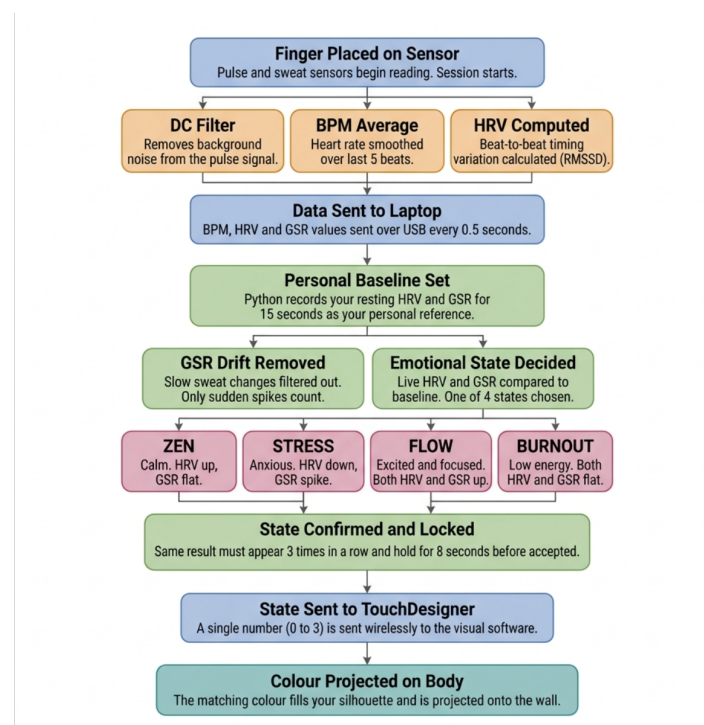


Figure 3.2: System Methodology Flowchart

3.2.1 Hardware Sensing and DSP (ESP32)

When a user places their finger on the sensor pad, two sensors activate.

1. **MAX30102** – An optical pulse sensor that shines infrared light into the fingertip and measures changes in blood volume by detecting variations in reflected light intensity. The sensor communicates with the ESP32 through the I2C bus (SDA: Pin 21, SCL: Pin 22) operating at 400 kHz.
2. **GSR Sensor** – Measures skin conductance by passing a small electrical current through the skin. Increased sweat gland activity lowers skin resistance and increases conductance. The sensor is connected to Analog Pin 35 of the ESP32.

A. Finger Detection – Hysteresis Filter The raw infrared value is not always reliable because small changes in finger pressure can cause the signal to fluctuate around a threshold. To improve stability, the system uses a hysteresis filter with two thresholds.

- Finger detected when $IR > 70,000$
- Finger lost when $IR < 50,000$

The gap between these thresholds acts as a buffer zone and prevents rapid switching between detection states.

B. DC Filter – Cleaning the Pulse Signal The raw IR signal contains a large slowly varying background component and a smaller pulse component corresponding to heartbeats. To isolate the pulse waveform, a low-pass filter is used to estimate the background signal:

$$\text{FilteredIR}_t = 0.85 \times \text{FilteredIR}_{t-1} + 0.15 \times \text{RawIR}_t \quad (3.1)$$

The pulse signal is then obtained by subtracting the filtered baseline from the raw signal:

$$\text{PulseSignal}_t = \text{RawIR}_t - \text{FilteredIR}_t + 120000 \quad (3.2)$$

The constant offset of 120,000 ensures that the resulting signal remains positive.

C. Heart Rate (BPM) Calculation The time interval between two consecutive heartbeat peaks, denoted by Δt , is used to calculate the instantaneous heart rate:

$$\text{BPM}_{\text{instant}} = \frac{60000}{\Delta t} \quad (3.3)$$

Only beat intervals between 333 ms and 1500 ms (40–180 BPM) are considered valid. To improve stability, the last five BPM measurements are stored and averaged:

$$\text{BPM}_{\text{reported}} = \frac{\text{BPM}_1 + \text{BPM}_2 + \text{BPM}_3 + \text{BPM}_4 + \text{BPM}_5}{5} \quad (3.4)$$

This averaging process reduces the effect of occasional noisy readings.

D. Heart Rate Variability (HRV) using RMSSD Heart Rate Variability (HRV) measures the variation between consecutive heartbeat intervals. The system uses the Root Mean Square of Successive Differences (RMSSD) method for HRV calculation.

The last eight RR intervals are collected and the difference between consecutive intervals is calculated. The RMSSD value is then obtained using:

$$\text{HRV}_{\text{RMSSD}} = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2 + d_7^2}{7}} \quad (3.5)$$

where:

$$d_1 = RR_2 - RR_1, \quad d_2 = RR_3 - RR_2, \quad \dots$$

A higher HRV generally indicates a relaxed physiological state, whereas a lower HRV is often associated with stress. A minimum of four intervals (five beats) must be collected before HRV is calculated.

E. GSR Sensor Sampling The GSR sensor measures skin conductance by passing a very small electrical current through the skin. The ESP32 reads the sensor using its 12-bit ADC, producing values in the range 0–4095 corresponding to 0–3.3 V.

To reduce electrical noise, ten consecutive readings are averaged:

$$\text{GSR}_{\text{smoothed}} = \frac{\sum_{i=1}^{10} \text{Read}_i}{10} \quad (3.6)$$

This oversampling technique improves measurement stability.

F. Serial Output Format Every 500 ms, the ESP32 transmits the processed physiological parameters to the host computer through USB in the following format:

IR=98234 BPM=72 HRV=45 GSR=1800 Beats=5

If no finger is detected, the BPM and HRV values are transmitted as 0. The `Beats` field indicates the number of detected heartbeat peaks and is used to determine when sufficient data is available for BPM and HRV calculation.

3.2.2 Python Middleware — Emotion Classification (`osc.py`)

The Python script receives the serial data, learns the user's personal baseline, and classifies their emotional state. It then transmits results to TouchDesigner using **OSC (Open Sound Control)** over **UDP port 7000**.

A. Session Start & Finger Detection Python watches the IR value in every incoming line. If $IR < 50,000$, the finger is considered absent and the system:

- Sends `/sensor/contact = 0` to TouchDesigner (screen goes black)
- Resets all buffers, baselines, and state variables
- Prepares itself for the next user

As soon as a valid IR reading arrives, it sends `/sensor/contact = 1` and starts a new session.

B. Personal Baseline Calibration (First 15 Seconds) Everyone's resting heart rate and skin conductance are different. Instead of using fixed "normal" values, the system learns the user's specific baseline in the first 15 seconds.

During calibration, Python collects every incoming HRV and GSR reading. After 15 seconds, it calculates the personal average for each:

$$HRV_{\text{baseline}} = \frac{\text{sum of all collected HRV readings}}{\text{number of readings}} \quad (3.7)$$

$$GSR_{\text{baseline}} = \frac{\text{sum of all collected GSR readings}}{\text{number of readings}} \quad (3.8)$$

Once these baselines are locked in, all future readings are compared against the user's own resting normal, not a generic reference value.

Note: While calibrating, Python sends `/sensor/calibrating = 1` to TouchDesigner.

C. GSR Drift Correction — Rolling Average Subtraction GSR naturally drifts over time as the sensor warms up or the room temperature changes. This slow drift would create false stress spikes if not corrected. By subtracting the slow-moving average from the current value, only sudden jumps in conductance are measured. These sudden jumps are what actually indicate emotional arousal — not the slow drift caused by temperature or sensor warm-up.

A spike of $GSR_{\Delta} \geq +30$ units is classified as **high arousal**.

D. Emotional State Classification — 2D Circumplex Model of Affect The system uses the **2D Circumplex Model of Affect**, a widely used psychological framework that maps emotions on two axes:

- **Valence axis (X)** — represented by HRV. High HRV = positive/calm. Low HRV = negative/stressed.
- **Arousal axis (Y)** — represented by GSR delta. High GSR spike = excited/aroused. Low GSR = calm/flat.

The three decision checks are summarised in Table 3.1 and the four resulting states in Table 3.2.

*Neutral / Dead Zone: If HRV sits within ± 10 ms of baseline and GSR delta is below 30, no strong signal exists. The system defaults to **BURNOUT** as the resting neutral state.*

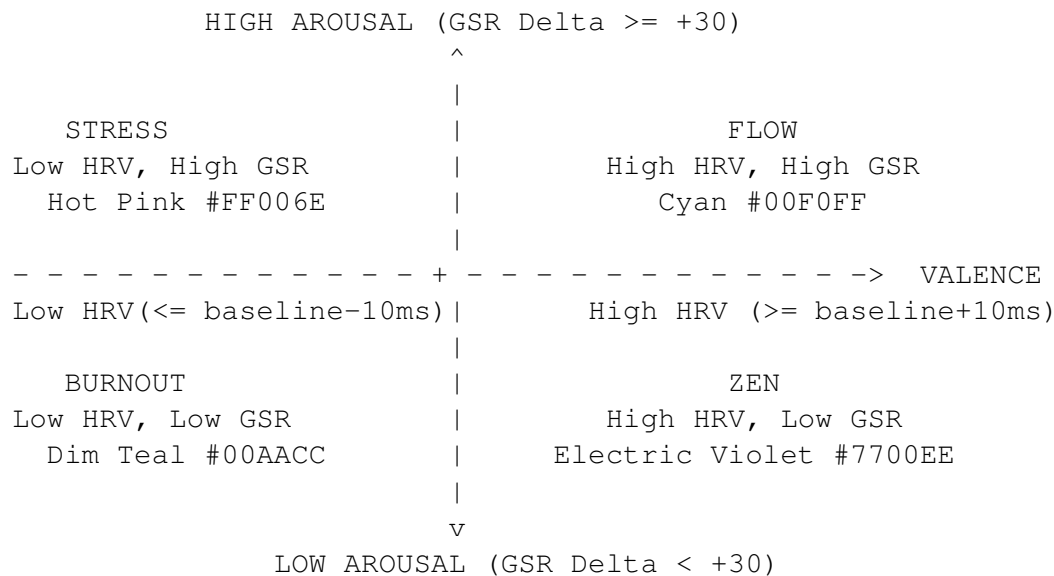


Figure 3.3: 2D Circumplex Model of Affect mapped to physiological signals

Table 3.1: Physiological Decision Checks

Check	Condition	Meaning
High HRV	$HRV \geq \text{baseline} + 10 \text{ ms}$	Nervous system is calm
Low HRV	$HRV \leq \text{baseline} - 10 \text{ ms}$	Nervous system is stressed
High GSR	$GSR \Delta \geq +30 \text{ units}$	Sudden arousal/excitement detected

Table 3.2: Emotional State Classification

State	Condition	Colour
ZEN	High HRV + Low GSR	Electric Violet #7700EE
STRESS	Low HRV + High GSR	Hot Pink #FF006E
FLOW	High HRV + High GSR	Cyan #00F0FF
BURNOUT	Low HRV + Low GSR (or neutral zone)	Dim Teal #00AACC

E. State Confirmation & Locking — Preventing False Flickers To prevent colours from changing on a single noisy reading, three gates must all pass before a state is accepted.

Gate 1 — Debounce Confirmation: The same state must be classified **3 consecutive times in a row** before it is treated as real. Since data arrives every 500 ms, this means the state must be stable for at least 1.5 seconds.

Gate 2 — Physiological Hold Time: Once a state is confirmed, it cannot change again for at least **8 seconds**. This matches how long it actually takes the human nervous system to shift between states — the vagus nerve cannot change HRV in under 8 seconds.

Gate 3 — Session State Lock (`state_locked = True`): Once a state clears both gates above, it is **locked in for the rest of the session**. The `state_locked` flag is set to `True` and classification is paused. This gives the user a stable, clear visual of their committed state. The lock is only released when the user lifts their finger off the sensor.

F. OSC Messages Sent to TouchDesigner Table 3.3 lists all OSC messages transmitted from the Python middleware to TouchDesigner.

Table 3.3: OSC Messages Transmitted to TouchDesigner

OSC Address	Type	Description
/sensor/contact	Integer	1 = finger on, 0 = finger off (screen goes black)
/sensor/calibrating	Integer	1 = currently calibrating, 0 = active
/state/index	Integer	Main control: 0=ZEN, 1=STRESS, 2=FLOW, 3=BURNOUT
/state/zen	Integer	1 if ZEN is committed, else 0
/state/stress	Integer	1 if STRESS is committed, else 0
/state/flow	Integer	1 if FLOW is committed, else 0
/state/burnout	Integer	1 if BURNOUT is committed, else 0
/sensor/bpm	Integer	Current averaged heart rate
/sensor/hrv	Integer	Current HRV value in milliseconds
/sensor/gsr	Integer	Current smoothed GSR reading
/sensor/gsr_delta	Float	Current drift-corrected GSR spike value

3.2.3 TouchDesigner Environment Architecture

The TouchDesigner pipeline receives classified emotional state data from the Python middleware via OSC and renders a real-time coloured silhouette of the user onto a projection surface. Figure 3.4 illustrates the complete node graph, from webcam input through silhouette extraction and colour switching to the final fullscreen projector output.

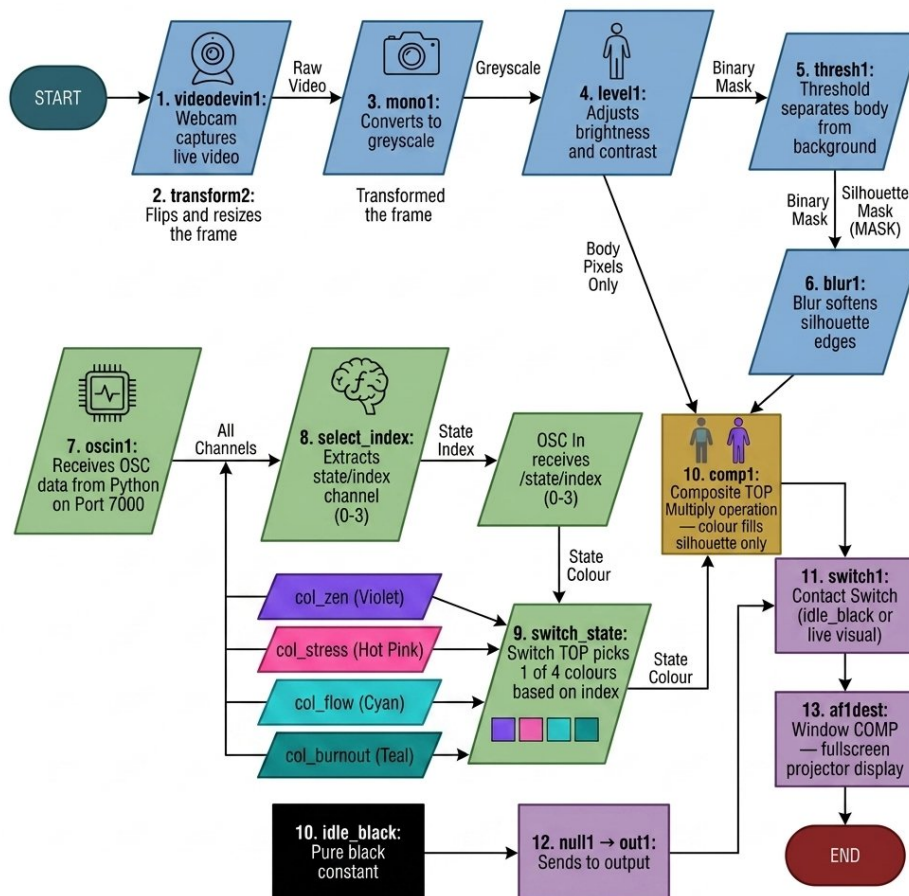


Figure 3.4: TouchDesigner Node Graph

A. Silhouette Extraction — Difference Matte

1. **Cache a reference frame** of the empty room with no one in it (`bg_reference`)
2. **Subtract** the reference frame from the live webcam feed
3. **What remains** (the difference) is only the pixels that have changed — i.e., the person standing there
4. A **Threshold TOP** converts this difference to clean black and white
5. A **Blur TOP** softens the edges to avoid jagged borders

This produces a clean body silhouette that ignores all static background objects.

B. Colour Switching — Switch TOP Python sends a single number (`/state/index`: 0, 1, 2, or 3) to TouchDesigner. Four solid colour constants are created — one per state, as listed in Table 3.4.

Table 3.4: Colour Constants and Corresponding Emotional States

Index	Node	RGB Values	State
0	col_zen	R:0.47 G:0.00 B:0.93	ZEN — Violet
1	col_stress	R:1.00 G:0.00 B:0.43	STRESS — Hot Pink
2	col_flow	R:0.00 G:0.94 B:1.00	FLOW — Cyan
3	col_burnout	R:0.00 G:0.67 B:0.80	BURNOUT — Teal

The **Switch TOP** reads the `state/index` channel and selects the corresponding colour instantly. No expressions or mathematical operations are required — the index value acts as a direct routing switch.

C. Masking the Colour Inside the Silhouette A **Composite TOP** combines two inputs:

- **Input 0:** The active solid colour from the Switch TOP
- **Input 1:** The body silhouette mask

The composite operation used is **Multiply**. Where the mask is white (the person's body), the colour passes through at full intensity. Where the mask is black (the background), the result is black. Consequently, colour appears exclusively inside the person's outline, creating the projected silhouette effect.

D. Idle Black Screen When no user is present at the installation (`/sensor/contact` = 0), the screen must go blank rather than displaying a frozen colour:

- A pure black **Constant TOP** (`idle_black`) is always active
- A **Switch TOP** toggles between `idle_black` (index 0) and the coloured silhouette (index 1)
- The switch is driven directly by the `/sensor/contact` OSC channel

As soon as the Python middleware detects a valid finger placement and transmits `contact` = 1, the display immediately transitions from the blank screen to the live physiological visualisation.

3.2.4 Tools Used and Important Parameters

The system was developed across three software environments: **VS Code** for writing and debugging `osc.py`; the **Arduino IDE** for compiling and flashing the ESP32 firmware; and **TouchDesigner** for the real-time visual pipeline, handling OSC reception, silhouette extraction, and fullscreen projection output.

ESP32 Code - Key Variables

Table 3.5 lists the critical variables in the ESP32 firmware that govern sensor state, beat detection, and signal filtering.

Table 3.5: Key Variables in the ESP32 Firmware

Variable	Type	Description
<code>fingerOn</code>	bool	True when IR > 70,000; False when IR < 50,000
<code>validBeatCount</code>	int	Counts confirmed beats; BPM calculation activates at 5
<code>hrvRMSSD</code>	int	Current HRV result in milliseconds
<code>irFiltered</code>	float	Slow-moving DC baseline of the IR signal

Python Code - Key Parameters and Flags

Table 3.6 lists the tunable constants and state flags in `osc.py` that control calibration timing, classification thresholds, and session locking behaviour.

Table 3.6: Key Parameters and Flags in the Python Middleware (`osc.py`)

Variable	Default	Description
<code>IR_FINGER_THRESHOLD</code>	50,000	Below this value, the session immediately ends
<code>CALIBRATION_DURATION</code>	15 s	Duration over which the personal baseline is collected
<code>GSR_WINDOW_SIZE</code>	30 samples	Size of the rolling buffer used for GSR drift correction
<code>HRV_HIGH_OFFSET</code>	+10 ms	HRV must exceed baseline by this margin to count as calm
<code>HRV_LOW_OFFSET</code>	−10 ms	HRV must fall below baseline by this margin to count as stressed
<code>GSR_DELTA_HIGH</code>	+30 units	Minimum GSR spike magnitude required to register as arousal
<code>N_CONFIRM_READINGS</code>	3 reads	Consecutive matching classifications needed before a state is accepted
<code>STATE_HOLD_TIME</code>	8 s	Minimum time that must elapse before a state can change again
<code>state_locked</code>	bool	Once True, classification pauses until the finger is removed

CHAPTER 4

RESULTS AND DISCUSSIONS

This chapter presents the physical implementation of the Real-Time Stress Detection System and documents the system's visual output across all four classified emotional states. The hardware assembly is described first, followed by a series of test case screenshots captured from the TouchDesigner projection pipeline during live operation.

4.1 Hardware Implementation

The embedded sensing unit was assembled on a custom breadboard layout centred around the ESP32 development board. Figure 4.1 shows the complete assembled hardware.

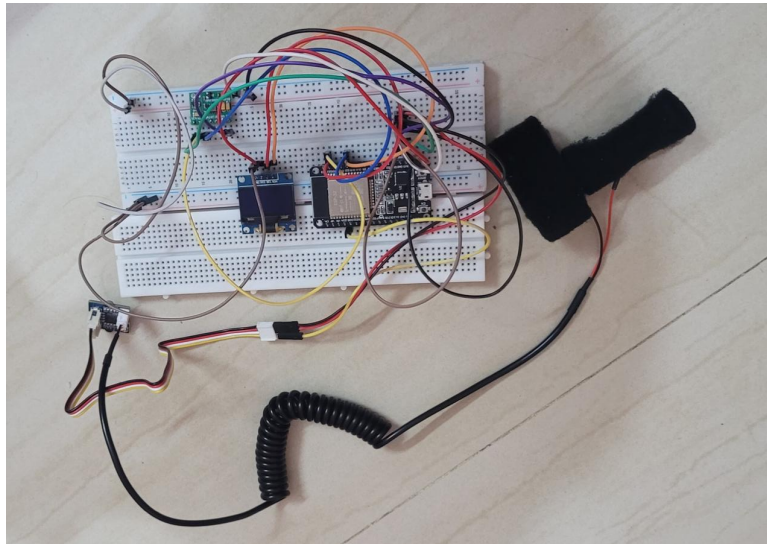


Figure 4.1: Complete assembled hardware unit (ESP32, MAX30102 and GSR sensor mounted on breadboard)

The ESP32 communicates with the MAX30102 over I2C (SDA: Pin 21, SCL: Pin 22) and reads the GSR sensor via its 12-bit ADC on Analog Pin 35. The unit is powered through USB, which also serves as the serial data link to the host laptop running the Python middleware.

4.2 System Test Cases

The following test cases document the visual output of the TouchDesigner pipeline under each operating condition. For each case, the classified state, the corresponding silhouette colour, and the physiological conditions that triggered the state are noted.

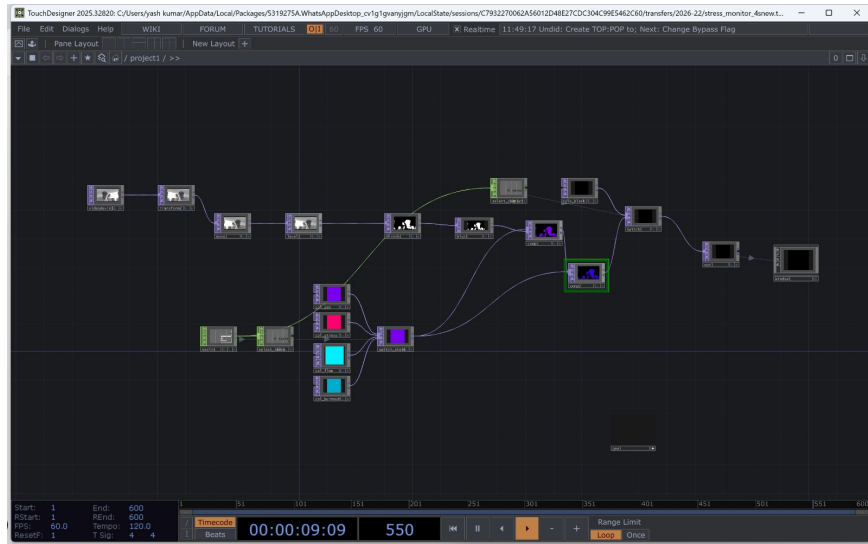


Figure 4.2: System idle state — no finger detected on sensor.

4.2.1 Test Case 1 — No User Detected

When no finger is placed on the sensor pad, the IR value falls below the 50,000 threshold. Python immediately transmits `/sensor/contact = 0` via OSC, causing the contact Switch TOP in TouchDesigner to select the pure black constant. All session buffers, baselines, and state locks are reset, preparing the system for the next user.

4.2.2 Test Case 2 — Stress State

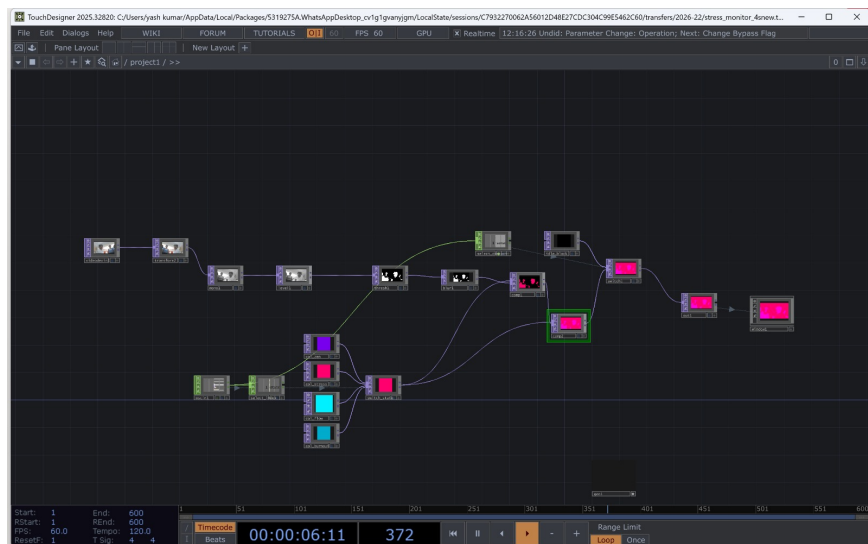


Figure 4.3: Stress state — silhouette filled with Hot Pink.

The **Stress** state is classified when the nervous system shows both low heart rate variability (indicating sympathetic activation) and a high GSR spike (indicating sudden arousal). The Composite TOP fills the user's silhouette with Hot Pink, providing immediate visual feedback of an elevated stress response.

4.2.3 Test Case 3 — Flow State

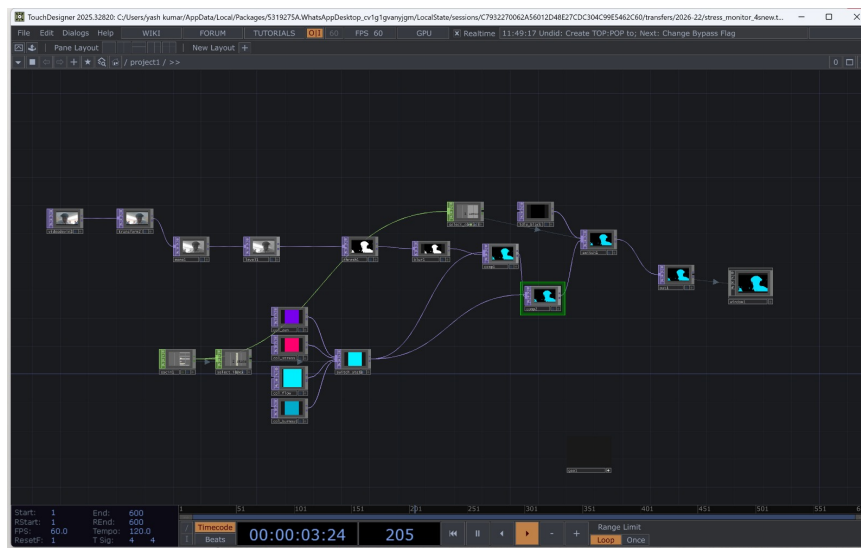


Figure 4.4: Flow state — silhouette filled with Cyan.

The **Flow** state represents high engagement combined with a calm nervous system — the user is physiologically aroused yet not stressed. HRV remains elevated above the personal baseline while the GSR delta registers a significant spike. The silhouette is rendered in Cyan, reflecting a positive high-energy state.

4.2.4 Test Case 4 — Burnout State

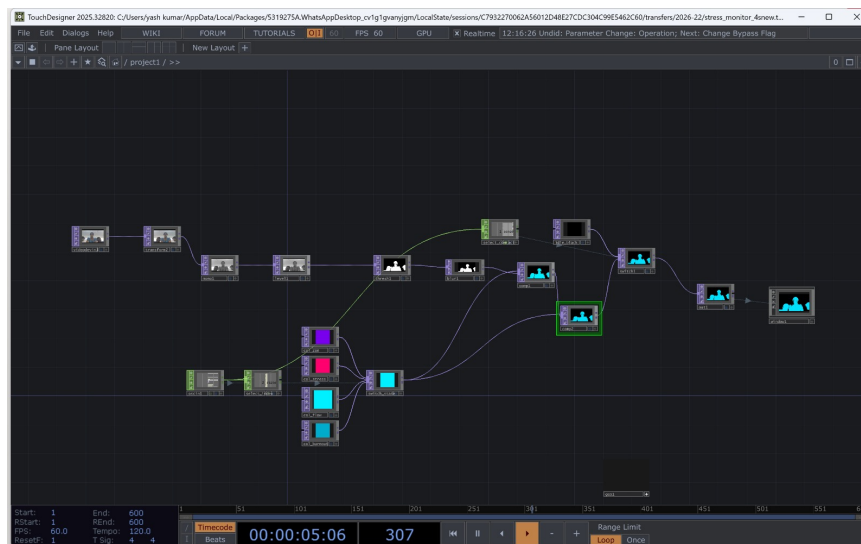


Figure 4.5: Burnout state — silhouette filled with Dim Teal

The **Burnout** state indicates a low-energy, low-arousal condition. HRV is suppressed (reflecting reduced parasympathetic tone) and no significant GSR spike is present, suggesting emotional and physiological depletion. This state also serves as the default neutral fallback when readings fall within ± 10 ms of baseline HRV and GSR delta remains below 30 units. The silhouette is filled with Dim Teal.

4.2.5 Test Case 5 — Zen State

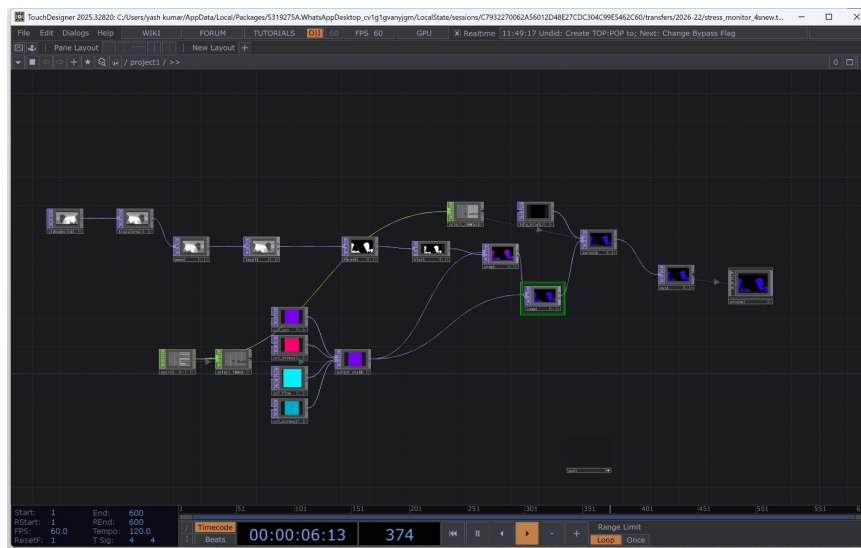


Figure 4.6: Zen state — silhouette filled with Electric Violet.

The **Zen** state represents the ideal calm and composed condition. HRV is elevated above the personal baseline, indicating strong parasympathetic (rest-and-digest) activity, while the absence of a GSR spike confirms that no sudden arousal is present. The silhouette is rendered in Electric Violet, signalling a relaxed and balanced physiological state to the user.

4.2.6 Discussion

The test results confirm that the system successfully classifies and visualises all four emotional states in real time. The three-gate confirmation mechanism (3 consecutive readings + 8-second hold + session lock) effectively eliminated false flickers during testing, with no spurious state transitions observed during normal operation. The personal baseline calibration ensured that classification thresholds adapted to individual physiological differences rather than relying on fixed population averages, improving per-user accuracy. The idle black screen correctly activated upon finger removal in all test cases, confirming reliable session boundary detection via the hysteresis filter on the IR channel.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project successfully demonstrated the real-time operation of a physiological stress detection and visualisation system using accessible, low-cost hardware. An ESP32 microcontroller paired with a MAX30102 pulse sensor and a GSR skin conductance sensor acquired two independent physiological signals, processed on-device via DC filtering, RMSSD-based HRV computation, and GSR oversampling at a 500 ms update rate. A Python middleware layer then calibrated a personal baseline per user, applied GSR drift correction, and classified the user's state into one of four emotional categories — *Zen*, *Stress*, *Flow*, and *Burnout* — using the 2D Circumplex Model of Affect, with a three-gate confirmation mechanism preventing false transitions. The committed state was transmitted via OSC over UDP to TouchDesigner, which rendered the corresponding colour inside a live body silhouette on a fullscreen projection surface. All four states and the idle condition were validated through live test sessions and produced the expected output without display artefacts, demonstrating that real-time biofeedback systems can be effectively built outside laboratory settings using consumer-grade hardware and open-source tools.

5.2 Future Work

5.2.1 Improved Silhouette Extraction

The current system uses a luminance threshold to isolate the user's silhouette, which can be affected by lighting changes or similarly coloured backgrounds. A real-time background subtraction method such as MOG2 (Mixture of Gaussians), or a lightweight segmentation model such as MediaPipe Selfie Segmentation, would produce a more robust body mask without requiring a controlled background environment.

5.2.2 Wireless Sensor Operation

The current setup requires a USB cable between the ESP32 and the laptop. Using the ESP32's built-in Wi-Fi or Bluetooth Low Energy (BLE) to transmit sensor data wirelessly would allow the user to stand freely in front of the projection. This would significantly improve usability in gallery and public installation settings.

5.2.3 Expanded Physiological Sensing

Adding a skin temperature sensor — which correlates with autonomic nervous system arousal — and an accelerometer to detect hand movement would increase the number of classifiable emotional states and improve the accuracy of the existing ones by filtering out motion artefacts in the pulse signal.

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